

Tranexamic Acid for Severe Hemorrhage in Trauma Patients

Guideline Title: Tranexamic Acid for Severe Hemorrhage in Trauma Patients	Guideline Number: GUID-3203-T016
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Issue date: 06/06/2016	Replaces Dept. Policy: GUID-3203-5016
Revision dates: 6/8/2021	Developed by: Air Care Team
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XXVIII. **PURPOSE:** This department guideline outlines the administration of Tranexamic Acid (TXA) by the Air Care Team for trauma patients, with indications of severe hemorrhage from blunt or penetrating trauma. TXA is an antifibrinolytic agent that blocks plasminogen from converting to plasmin, an enzyme that dissolves blood clots. It works to stabilize clots that have formed at the site of injury.

XXIX. **DEFINITIONS:**

- a. TXA - Tranexamic Acid
- b. SBP – Systolic blood pressure

XXX. **DEPARTMENT GUIDELINE:**

- a. **Indications:** TXA is given to trauma patients with suspected severe hemorrhagic shock as a result of a penetrating or blunt injury as evidence by:
 - i. **Adult patients** and patients 16 years old and older
 1. Sustained SBP < 90mmHg **AND**
 2. Sustained heart rate of greater than 110 beats per minute **OR**
 3. Obvious signs of sustained hemorrhagic shock after 1000ml of crystalloid infusion **OR**
 4. Initiation of *Massive Transfusion Protocol* at receiving facility
 - ii. **Pediatric patients** 15 years and younger
 1. Sustained SBP of <70 + (age in years x 2) for children <10 years old and SBP <90 for children >10 years old **AND**
 2. Sustained tachycardia appropriate for clinical context and age **AND**
 - a. Neonates (less than 28 days) < 100bpm or > 180bpm
 - b. Infant (1 month – 1 year) < 80 bpm or > 160bpm
 - c. Child (1 – 10 years) < 60 bpm or > 130bpm
 - d. Older Child (>10 years) < 60 bpm or > 110bpm
 3. Obvious signs of sustained hemorrhagic shock and hypoperfusion (pallor, peripheral vasoconstriction including cool, pale skin and delayed capillary refill > 2 seconds) **after 20ml/kg of IV crystalloid infusion. Pediatric patients must receive IV fluid bolus prior to receiving TXA.**

- b. TXA must be started within 3 hours from time of injury with best results occurring if started within 1 hour from time of injury.
- c. **Contraindications:**
 - i. Known hypersensitivity
 - ii. History of known thromboembolic disease such as a deep vein thrombosis (DVT) or pulmonary embolism (PE).
 - iii. Pregnancy with fetus of viable gestational age
 - iv. Patients who have contraindication to anti-fibrinolytic therapy agents
- d. Establish care as defined by *GUID-G002 – General Patient Care* or *GUID-PED001 – Pediatric General Management*.
 - i. **Adult** patients and patients 16 years old and older
 - 1. Bolus of **1-gram TXA mixed in 100ml of Normal Saline (NS) IV/IO** over 10 minutes or 600ml/hour. Do not give IV push (more rapidly than 10 mL/min) due to potential for hypotension.
 - 2. Continue any maintenance infusion from a sending facility or consider a maintenance infusion if bolus was previously received: Typically, 1-gram TXA in 250ml or 500ml NS over 8 hours.
 - ii. **Pediatric** patients 15 years old and younger
 - 1. Bolus of **TXA 15mg/kg** mixed in appropriate volume of fluid with concentration of 10mg/ml of Normal Saline (NS) IV/IO (maximum 1 gram/100ml) over 10 minutes. Do not give IV push (more rapidly than 10 mL/min) due to potential for hypotension.
- e. Administration and patient care
 - i. If blood products are infusing, use an alternative IV site for TXA administration.
 - ii. Monitor for side effects: seizure, vision changes, renal impairment, and hypotension with rapid injection
 - iii. Do not delay transport; TXA may be given en route to trauma center/receiving facility.
 - iv. Provide a radio and verbal report to emergency department staff and trauma physician. Notify receiving facility of time the bolus was given and need to start infusion or time the bolus was given and the time the maintenance infusion was started.

XXXI. **DOCUMENTATION:**

- a. EMS Charts patient care record

XXXII. **REFERENCES:**

- a. CRASH-2 Trial Collaborators, Roberts et al., (2010) Effects of tranexamic acid on death, vascular occlusive events, and blood transfusion in traumatic patients with significant hemorrhage (CRASH-2): a randomized placebo-controlled trial. *Lancet* 376:23.
- b. Joint Theater Trauma System Clinical Practice Guideline for Damage-Control Resuscitation http://www.usaasr.amedd.army.mil/assets/cpgs/Damage_Control_Resuscitation_10_Aug_2011%20pdf
- c. Levy, J. H., Koster, A., Quinones, Q. J., Milling, T. J., & Key, N. S. (2018). Antifibrinolytic Therapy and Perioperative Considerations. *Anesthesiology*, 128(3), 657–670. <https://doi.org/10.1097/ALN.0000000000001997>
- d. Napolitano et al., (2013). Tranexamic acid in trauma: How should we use it? *J Trauma Acute Care Surg*, 74(6), 1575-1586.